PELA Ep. 2

2026-10-22

- Some changes in the stybos.
- The main merge of Ep. 2 & Ep. 3 combined:

Post-Manifold Hypothesis in Practice.

How to Approach Statements in Math:

1) (Attempt to) Assess the truth of the statement, either exactly or probabilistically.

2) Build a case. examples, draft of proof.

3) Construct a ~~proof~~ (formal) proof

4) Perturb the statement. or its proof.

fewer assumptions? Stronger conclusions?

I attempted def. in syllabus

- Post-Manifold Hypothesis: Edmund Husserl:

"Mathematics is a manifold in the sense phenomenological sense".

(Ep. 3)

- Some Naive Set Theory and Logic is Assumed. / See e.g. early lectures in my 3210.

(Truth Tables etc.

Proof Methods

Proof by Software / Tokens / Capital.

Proof by Wetware

Proof by (Flesh.)

Maurice Merleau-Ponty. (?).

Some Naive (Category Theory and) Categorical Logic:

Better reference):

- van Dalen - Logic and Structure
  Ch. 2, 3.
- Troelstra - Schwichtenberg -
  Basic Proof Theory
  Ch. 8.
- MacLane - Cats for the Working
  Mathematician.

  or any intro cat theory.

Set theory

↓

sets are like blobs.

Cat theory

↓

cats are like ~~graphs~~ flowcharts.

- How do cat. logic and std logic compare? (with truth tables)

They are fully compatible.

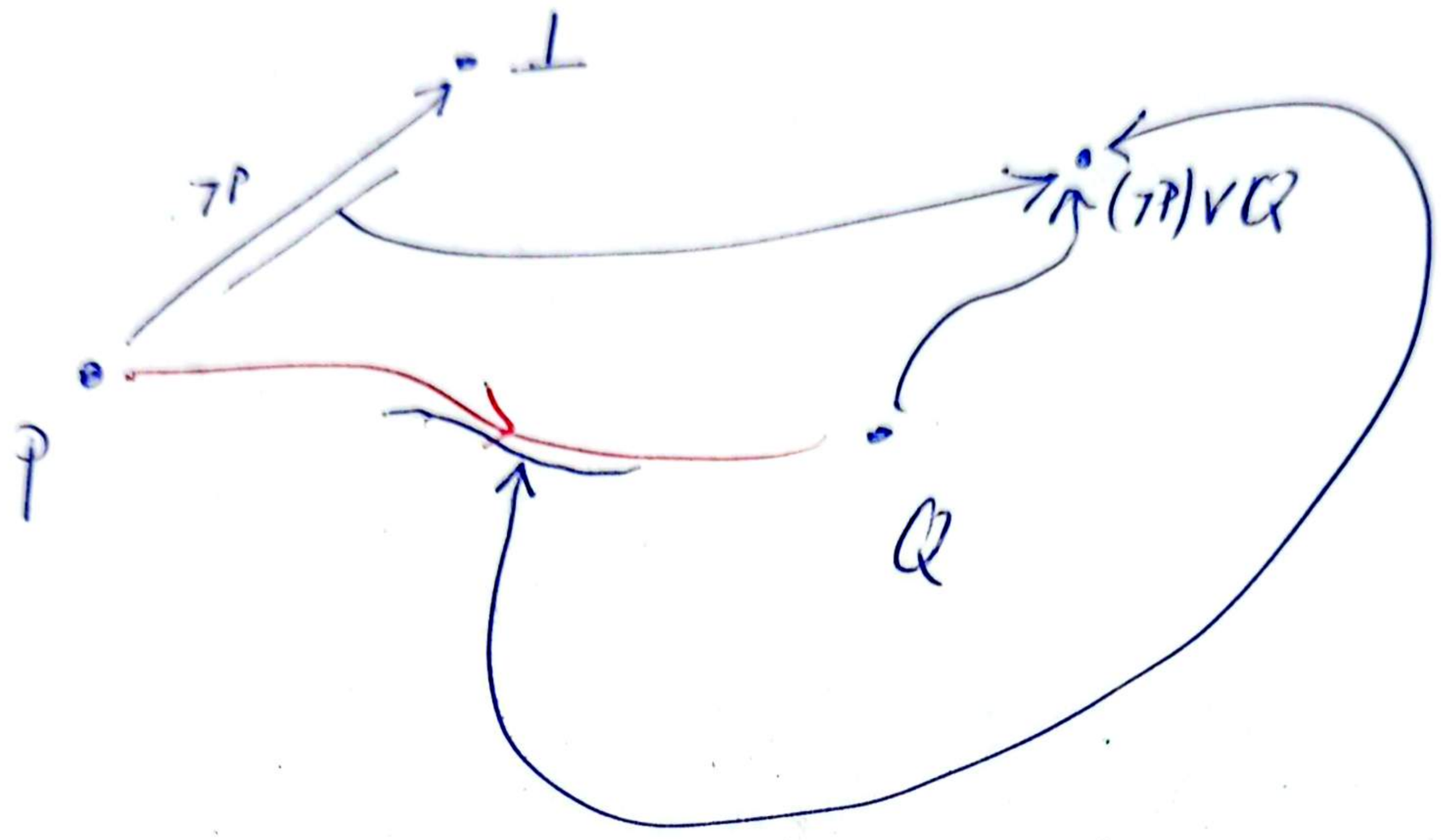
Truth tables are what preds. reduce to under observables (valuations/interpretations),

if they are about the semantics.

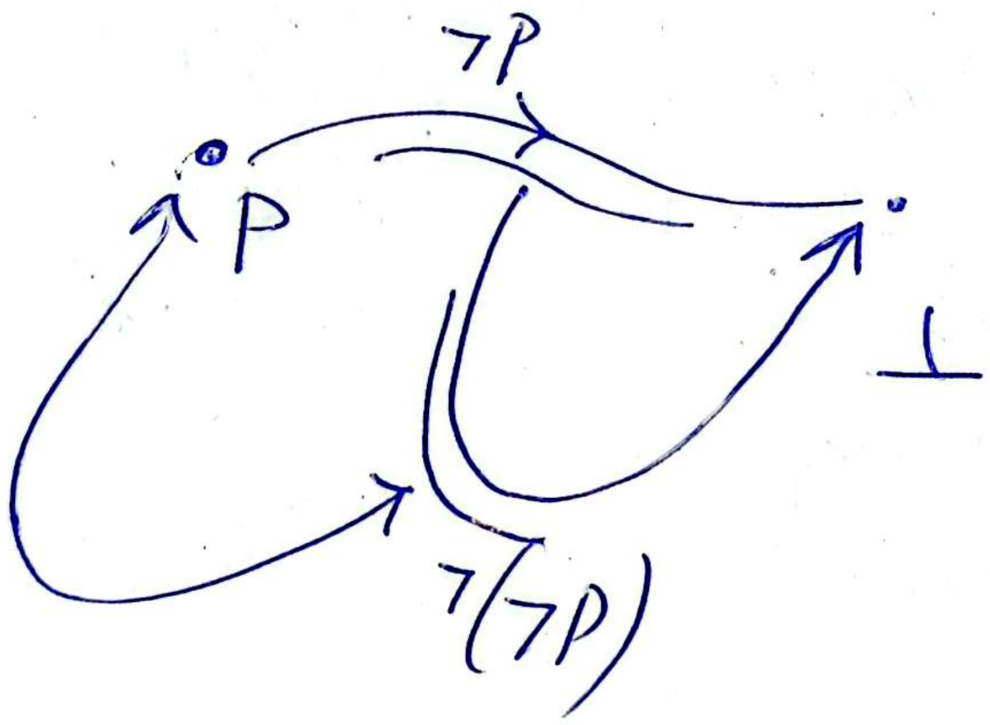
Cat. logic is more like dynamics in phase space. syntactic

↳ panta rhei (πάντα ῥεῖ).

Ex: $[P \to Q] \iff [(\neg P) \vee Q]$.



Ex: $[\neg(\neg P)] \iff P$



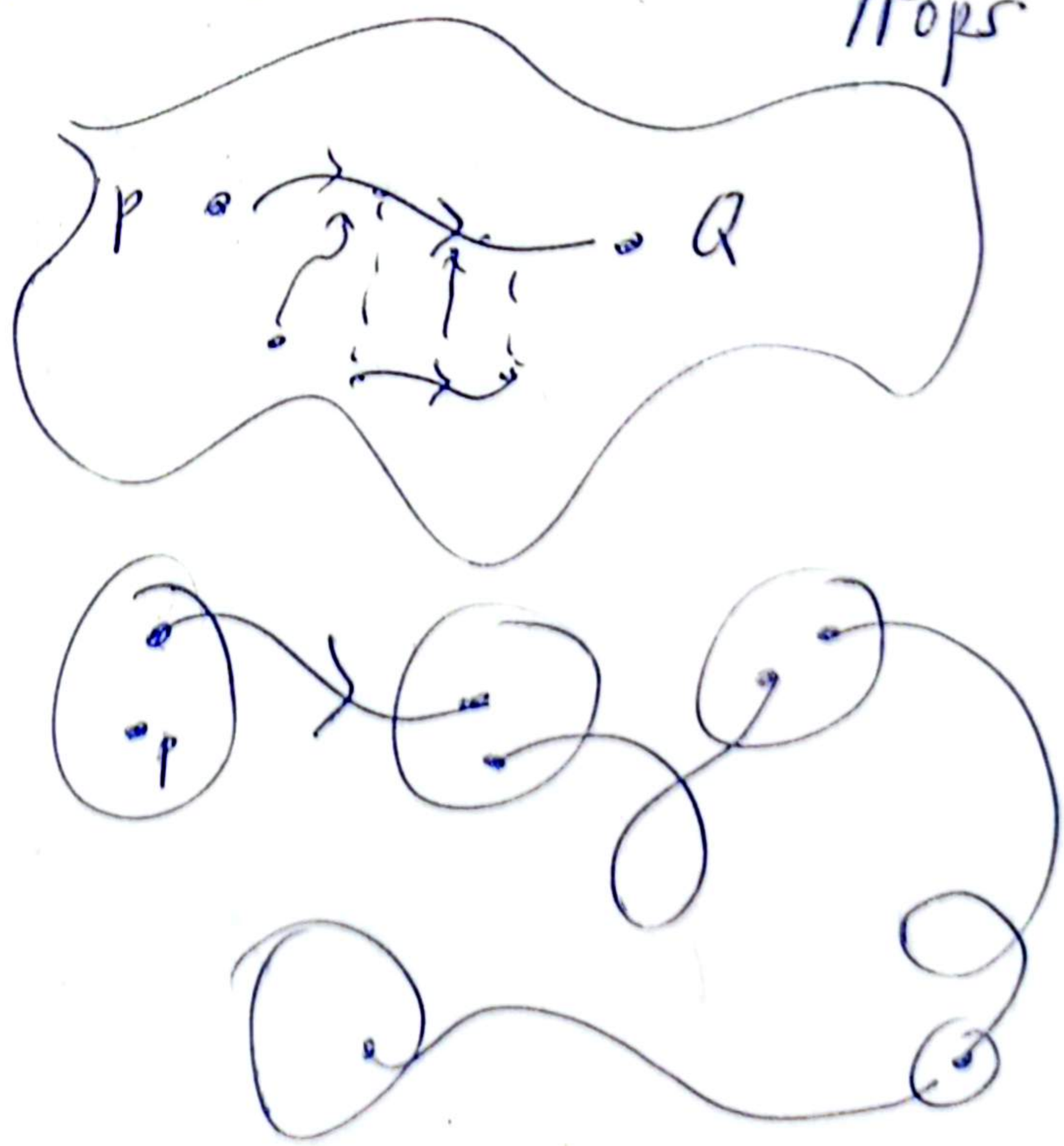
Layer 1: Propositions, Absurdum $\bot$, Verum T.
Connectives = Implies $\rightarrow$, And $\wedge$, Or $\vee$,
Not $\neg$, —

Layer 2: Predicates = parameterized families of Props

Logical Quantifiers - $\forall$ For all $\forall$
There exists $\exists$
There exists unique $\exists!$

Methods of Proof by Flesh:

- direct proof



H; T

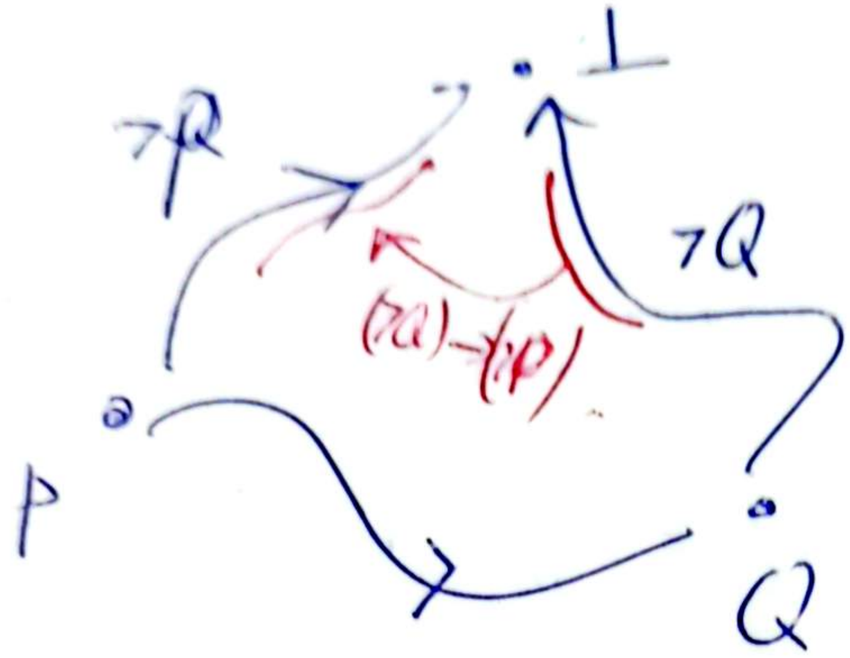
Props

AND

OR

- **proof by contrapositive**

$$[P \to Q] \iff [(\neg Q) \to (\neg P)].$$



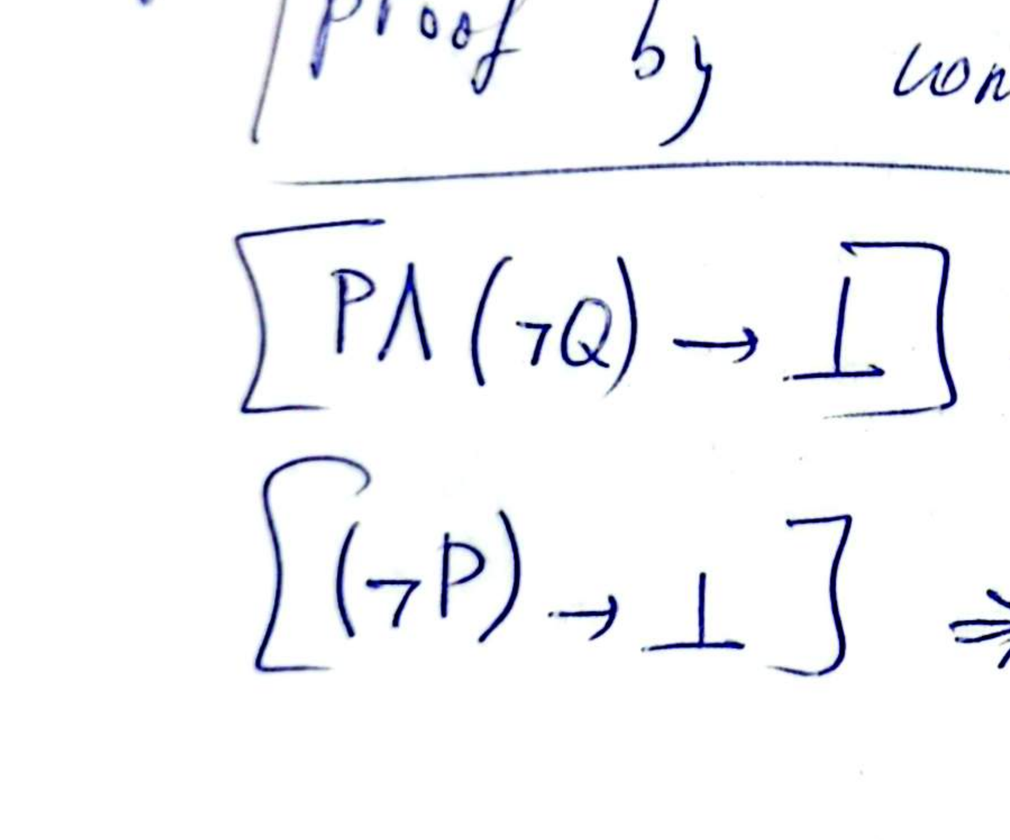
- **proof by contradiction.**

RAA
reductio ad absurdum.

$$[P \wedge (\neg Q) \to \bot] \Rightarrow [P \to Q].$$

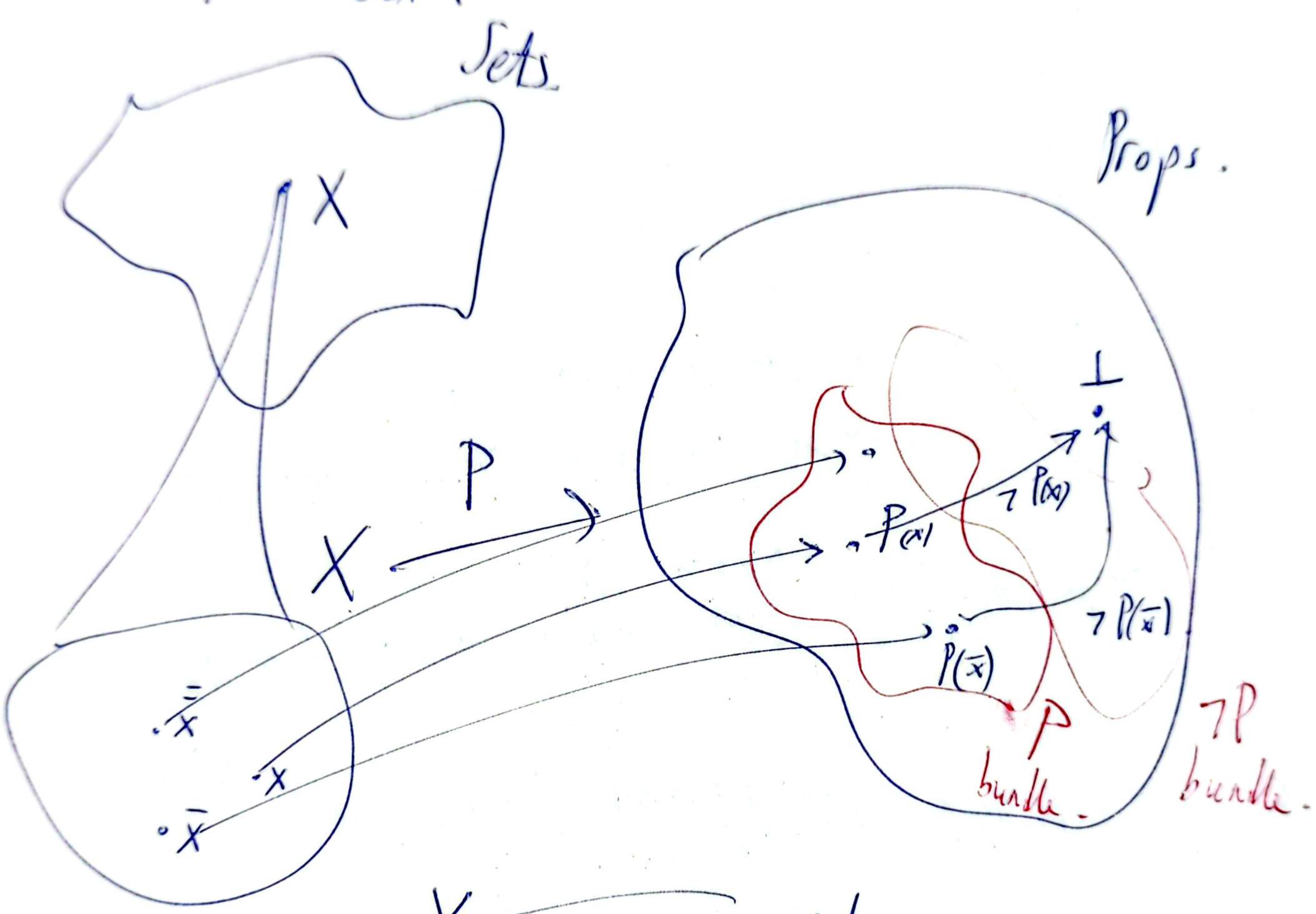
$$[(\neg P) \to \bot] \Rightarrow P.$$

P	¬P	(¬P) → ⊥
1	0	1
0	1	0



· Prop families. // Prop bundles.

X : set.



$$X_P = \{x \in X \mid v(P(x)) = 1\}$$

v valuation.

$$[\forall x \in X : P(x)] = [X_P = X]$$
$$[\exists x \in X : P(x)] = [X_P \neq \emptyset]$$
$$[\exists! x \in X : P(x)] = [\#(X_P) = 1]$$

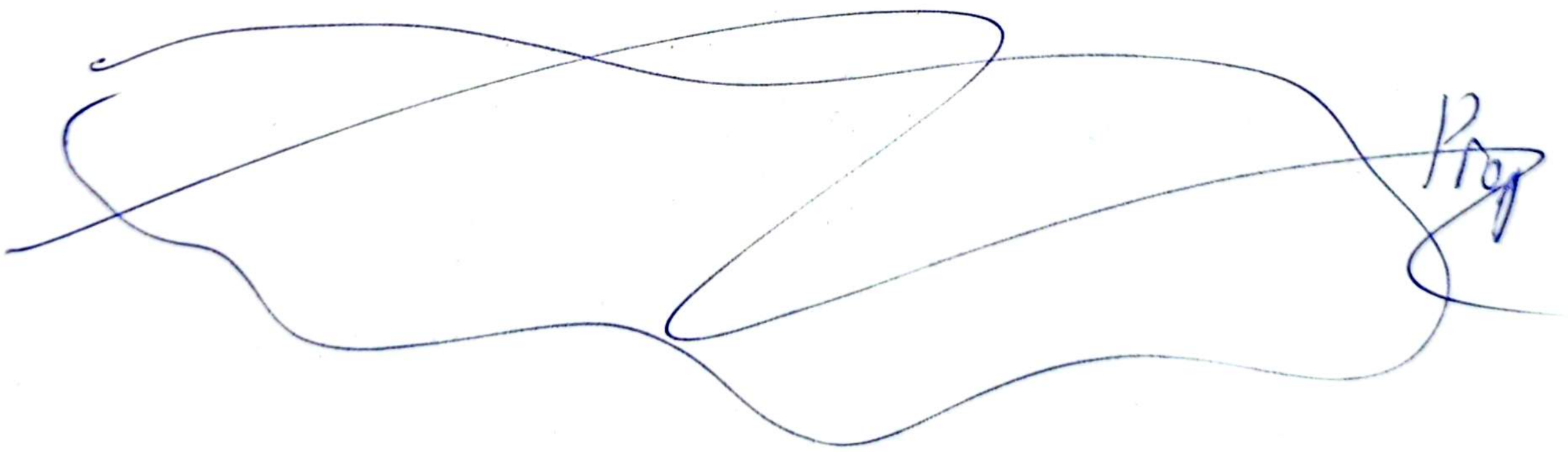
- [proof by induction]

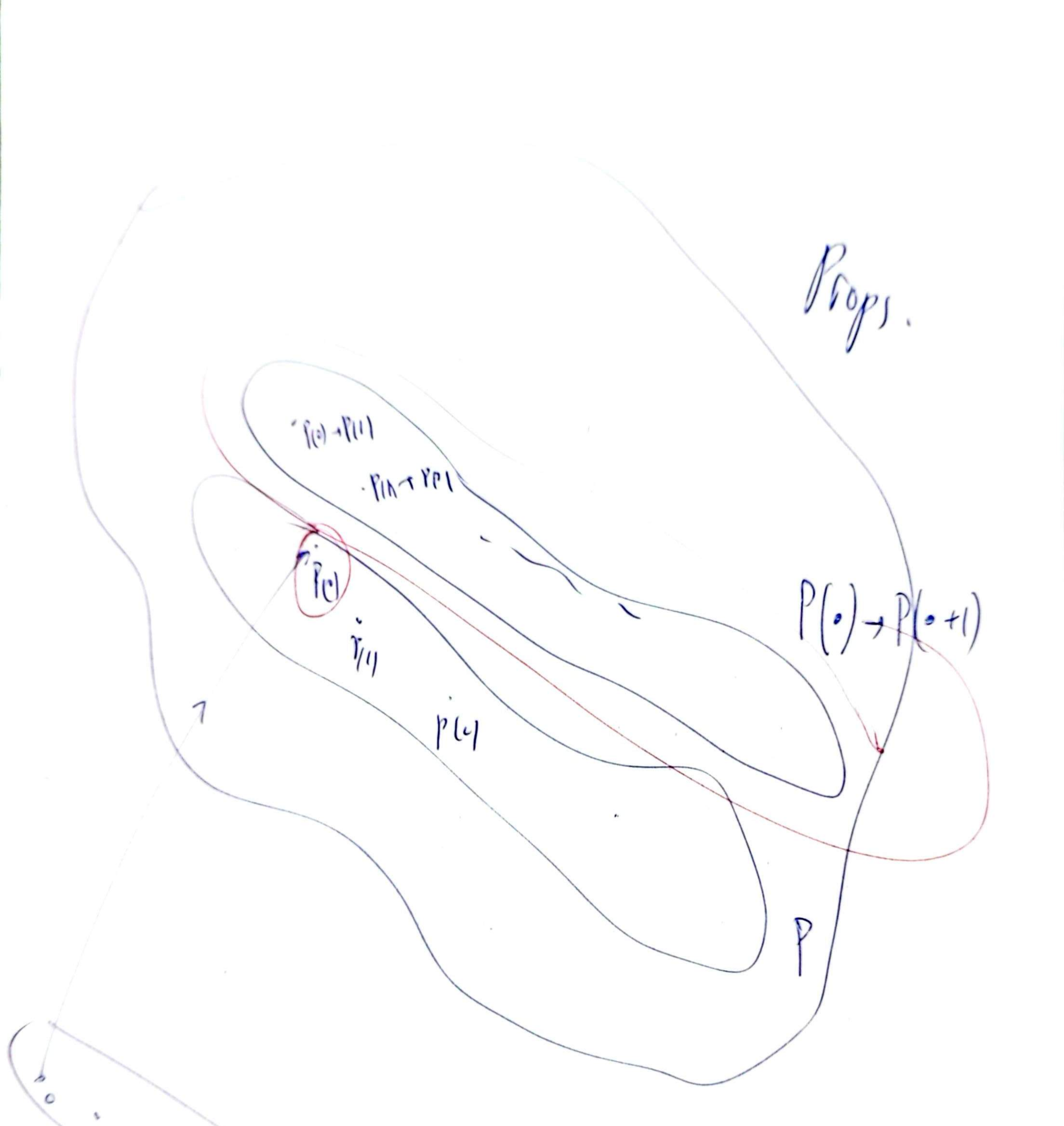
$X = \mathbb{Z}_{\geq 0}$. (more precisely X is the object of a net)

$P : \mathbb{Z}_{\geq 0} \to \mathrm{Props}$.

$[\forall n \in \mathbb{Z}_{\geq 0} : P(n)]$

$\Leftrightarrow [P(0) \wedge [\forall n \in \mathbb{Z}_{\geq 0} : P(n) \to P(n+1)]]$.




P(0) → P(1)
P(1)
P(0) → P(0+1)
P(0)
P